Math 161
Final Exam solutions

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1. If B has least upper bound $\sup B$, then in particular $\sup B$ is an upper bound for B. Since $A \subset B$, it follows that $\sup B$ is also an upper bound for A. We assumed that $A \neq \varnothing$, and hence by the least upper bound property $\sup A$ exists. Since $\sup B$ is an upper bound for A and $\sup A$ is the least upper bound, we have $\sup A \leq \sup B$ as claimed.

2. Let $M := \lim\limits_{y \to L} g(y)$ and fix $\epsilon > 0$, so we must show that there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies that $|g(f(x)) - M| < \epsilon$. Since $\lim\limits_{y \to L} g(y) = M$, there exists $\delta' > 0$ such that $0 < |y - L| < \delta'$ implies that $|g(y) - L| < \epsilon$. Similarly, the hypothesis $\lim\limits_{x \to a} f(x) = L$ implies that there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies that $|f(x) - L| < \delta'$. Moreover, if $|x - a| > 0$ then $x \neq a$ and hence $f(x) \neq L$, or equivalently $|f(x) - L| > 0$. Thus $0 < |x - a| < \delta$ implies that $0 < |f(x) - L| < \delta'$, which in turn implies that $|g(f(x)) - M| < \epsilon$ as desired.

3. If $f'(0)$ exists, then it is computed by the limit

$$f'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0} \frac{hg(h) - 0 \cdot g(0)}{h} = \lim_{h \to 0} g(h).$$

Since g is continuous at 0, the final limit exists and equals $g(0)$.

4. Note that f is differentiable everywhere, since

$$f'(x) = \frac{x^2 + 1 - x \cdot 2x}{(x^2+1)^2} = \frac{1 - x^2}{(x^2+1)^2}$$

by the quotient rule, and $(x^2+1)^2 > 0$ for all $x \in \mathbb{R}$. In particular f is continuous everywhere, and the extreme value theorem implies that f attains maximum and minimum values on $[0, 2]$. Moreover, we have seen in class that these global extrema must occur at either the endpoints $x = 0, 2$ or the critical points of f in $(0, 2)$ (in this case, there are no points where f fails to be differentiable). The equation $f'(x) = 0$ is equivalent to $1 - x^2 = 0$, which has the solutions $x = \pm 1$, so the only critical point of f in $(0, 2)$ is $x = 1$. We compute

$$f(0) = 0, \ f(1) = \frac{1}{2}, \ f(2) = \frac{2}{5}.$$

Thus f has minimum value $f(0) = 0$ and maximum value $f(1) = \frac{1}{2}$.

5. It is enough to prove that for any $x \in \mathbb{R}$ and any $\epsilon > 0$, we have $|f(x)| < \epsilon$. Namely, if this holds and $f(x) \neq 0$, then $|f(x)| > 0$ and hence $|f(x)| < |f(x)|$, a contradiction.

So fix $x \in \mathbb{R}$ and $\epsilon > 0$. Since f is continuous at x, there exists $\delta > 0$ such that $|x - y| < \delta$ implies that $|f(x) - f(y)| < \epsilon$. But A is dense in $\mathbb{R}$, so there exists $y \in A$ such that $x - \delta < y < x + \delta$, or equivalently $|x - y| < \delta$. Then we have

$$|f(x)| = |f(x) - f(y)| < \epsilon$$

as needed.